

EXPENSE MATE METRIX REPORT(24K-3075 , 24K-3028)

1. Size Metrics

Metric	Value
Production code (backend + screens + database)	2,806 LOC
Automated test code	841 LOC
Test-to-code ratio	≈ 0.30
Backend modules	14 files
Frontend/UI screens	13 files
Backend functions/methods analyzed	88

3. Code Complexity Metrics

3.1 Cyclomatic Complexity

Computed across all 88 backend functions and methods using radon.

Grade	Meaning	Count (of 88)
A (1–5)	Simple, low risk	78 (88.6%)
B (6–10)	Moderate complexity	9 (10.2%)
C (11–20)	Complex	0 (0%)
D (21–30)	High complexity — refactor candidate	1 (1.1%)

Average cyclomatic complexity: 2.83 (Grade A) — overall the codebase is low-risk.

3.2 Highest-Complexity Functions

Function	File	Grade (CC)
import_transactions_csv	backend/services/report_service.py	D (21)
update_transaction	backend/services/transaction_service.py	B (9)
add_transaction	backend/services/transaction_service.py	B (9)
verify_otp	backend/services/auth_service.py	B (8)
update_savings_goal	backend/services/savings_service.py	B (8)

get_report_transactions	backend/services/report_service.py	B (8)
register_user	backend/services/auth_service.py	B (7)
get_budget_status	backend/services/budget_service.py	B (7)

4. Test Metrics

4.1 Documented Test Case Summary

Category	Cases	Pass	Fail
Frontend unit tests (FT-01–40)	40	40	0
Backend unit tests (TC-01–38, representative)	38	38	0
Integration tests (IT-01–35)	35	35	0
Total documented cases	113	113 (100%)	0

5. Fault-Find & Fault-Fix Metrics

Faults were identified by executing the automated test suite against the repository and are tracked through to resolution below.

#	Fault Found	Fix Applied	Status
1	Hardcoded /tmp/ paths in test setup caused FileNotFoundError on Windows (9 of 16 report-service tests affected)	Replaced with tempfile.gettempdir() for OS-portable paths	Fixed
2	test_auth_service.py imported a class-based AuthService/AuthError API that did not exist in auth_service.py — entire module (14 tests) failed to load	Rewrote tests against the actual function-based API (register_user, login_user, verify_otp, etc.); coverage expanded 14 → 34 tests	Fixed

Metric	Value
Faults found	2
Faults fixed	2
Open faults	0
Fault-fix rate	100%
Defect density (faults / KLOC)	≈ 0.71 per 1,000 LOC

